

Solution

MAGNETISM - 05

Class 10 - Science

Section A

1. (a) 1-A, 2-C, 3-B, 4-D

Explanation:

- i. A lodestone is a naturally magnetized piece of the mineral magnetite. Loadstone is iron ore.
- ii. Magnetic flux is the number of magnetic field lines passing through a surface placed in a magnetic field.
- iii. An electric motor converts electrical energy into mechanical energy.
- iv. Permanent magnet are usually made of hard materials which are strongly magnetized. A perfect example of permanent magnet is the Bar Magnet. In fact, Permanent magnets are most of the times referred to as Bar Magnets also.

2.

- (c) 1-B, 2-D, 3-A, 4-C

Explanation: (1) The tesla (symbolized T) is the standard unit of magnetic flux density.

(2) A solenoid is a tightly wound helical coil of wire whose diameter is small compared to its length. The magnetic field generated in the centre, or core, of a current carrying solenoid is essentially uniform, and is directed along the axis of the solenoid.

(3) The needle of a magnetic compass is 'just' a small bar magnet that is balanced carefully so it can rotate freely.

(4) Solenoid is temporary strong magnet, when connected through external source like battery or when electricity passed through it. It is also known as electromagnet.

3.

- (c) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Explanation:

- Dynamo is a device that makes direct current electric power using electromagnetism. It is also known as a generator. Dynamos and generators work using the wild complex phenomena of electromagnetism.
- Magnetism is one aspect of the combined electromagnetic force. It refers to physical phenomena arising from the force caused by magnets, objects that produce fields that attract or repel other objects.
- A generator converts mechanical energy into electrical energy, while a motor converts electrical energy into mechanical energy. Both the devices work because of electromagnetic induction, which is when a voltage is induced by a changing magnetic field.
- Andre-Marie Ampere was a French physicist and mathematician who was one of the founders of the science of classical electromagnetism. The SI unit of measurement of electric current, the ampere, is named after him.

4.

- (c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Explanation:

- Green wire stands for grounding/ earthing in the electric circuit.
- A dry cell is a type of battery. It always produces a DC current.
- A solenoid is a coil wound into a tightly packed helix.
- An electromagnet is a temporary magnet that has a magnetic field produced by the flow of electric current in a coil wound around a soft iron core.

5.

- (d) 1-B, 2-D, 3-A, 4-C

Explanation: (1) Michael Faraday is credited with the discovery of electromagnetic induction on August 29, 1831.

(2) A current-carrying conductor, when placed in a magnetic field, experiences a force. if the direction of magnetic field and that of current are mutually perpendicular then force acting on the conductor will be perpendicular to both and will be the given by Fleming's left-hand rule.

(3) The direction of the magnetic field lines of force around a conductor is given by the Maxwell's right hand grip rule or the

right handed corkscrew rule.

(4) Hans Christian Oersted discover Electromagnetism.

6.

(d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- Like poles repel each other.
- Unlike poles attract each other.
- Live wire is red in colour.
- The neutral wire is black in colour.

Section B

7. **(a)** Straight

Explanation: The magnetic field lines inside the solenoid are in the form of parallel straight lines. The field lines inside the solenoid are in the form of straight lines which indicates that the magnetic field is the same at all points inside the solenoid.

8.

(c) Live wire and Neutral wire

Explanation: When the main switch in a building is turned off or the breaker trips or the fuse burns, power to both live and neutral lines is disconnected. The earth or ground line is a safety line that must never be disconnected.

9.

(b) 17°

Explanation: Magnetic axis is the line passing through earth's magnetic north and south poles which is inclined at an angle of nearly 17° with the geographic axis.

Section C

10.

(d) Both A and B

Explanation: It leads to the burning of the fuse wire because of the excessive flow of current, the temperature of the wire goes beyond the melting point of the fuse wire and hence it is melted by burning. So, A and B are the correct reasons.

11.

(c) Both the statement A and B are true

Explanation:

- A. Hans Christian Oersted, a Danish physicist first noticed the magnetic effect of electric current.
 - B. The strength of a magnetic field can be observed from the degree of closeness of the field lines. Where magnetic fields are closer magnetic strength is high as near poles of a magnet, and where these are far apart magnetic strength is weak.
- Statement (A) and (B) are true statements.

12.

(c) Both the statement A and B are true

Explanation:

- A. The magnetic field of a straight conductor is concentric around the conductor, so the field lines don't seem to start or end anywhere, so there are no poles.
- B. When two magnets are placed with their like poles facing each other, the lines of force are in opposite directions, and hence like poles repel each other.

Section D

13.

(d) All of these

Explanation: Overloading takes place when the live wire comes into direct contact with the neutral wire (This occurs when the insulation of wires is damaged or there is a fault in the appliance). In such a situation, the current in the circuit abruptly increases. This is called Short-circuiting. Overloading is caused by connecting too many appliances to a single socket. So, all statements are correct.

14.

(b) If magnetic field lines are parallel and equidistant, they represent zero-field strength

Explanation: This statement is false about the magnetic field lines that if magnetic field lines are parallel and equidistant then they represent zero-field strength because if they are parallel and are at an equal distance then they have a uniform magnetic field and don't have zero magnetic field strength.

15. **(a)** B and C

Explanation: A. A fuse wire is a safety wire connected in series with the live wire, in case of any large current supply or malfunctioning in the electric connections it melts and breaks the electric circuit.

B. A fuse wire is connected in series so as to pass the current totally through the fuse. It is not connected in the end of the circuit.

C. The relation between thickness of a wire and its current carrying capability is that, more the thickness more the current a wire can carry. So fuse wires are made of sufficient thickness. Means thinner the fuse wire, lesser is its current capacity.

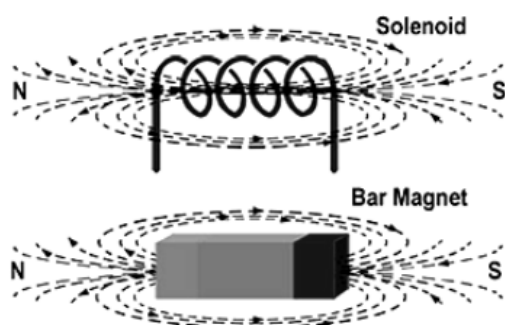
D. The fuse needs to be rated slightly higher than the maximum operating current of the protected device, so that the fuse won't blow when the equipment is used.

Statement B and C are not correct about electric fuse.

16. **(a)** The pattern of the magnetic field associated with the solenoid is different from the pattern of the magnetic field around a bar magnet

Explanation:

The pattern of the magnetic field associated with the solenoid is the same as the pattern of the magnetic field around a bar magnet.



17.

(c) A and B

Explanation:

A. A fuse is made of an alloy of tin and lead because it is important that the device acts without heating too much.

B. A switch in the electrical circuit is always connected to the live wire so that the socket/appliance is not live when it is switched off.

C. Alternating current is defined as the flow of charge that changes direction periodically.

D. Higher is the value of current flow, stronger is the magnetic field

So, Statement A and B are correct statement.

18. **(a)** D

Explanation: Magnetic field emerges from north pole to south pole and has only one direction, thus no two field lines overlap.

A wire with red insulation is usually the live wire of electric supply.

So, all statements are incorrect.

19. **(a)** Statement (c) is incorrect.

Explanation: The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet.

20.

(d) In India, the AC changes direction after every $\frac{1}{50}$ second

Explanation: The frequency of Alternating current in India = 50 cycles/second

Therefore, the time period of the alternating current $\frac{1}{50}$ second

Since AC changes the direction of current after every half time period. Therefore AC will change current in $\frac{1}{100}$ seconds.

21.

(d) A, B and C

Explanation:

- A. A curl is always present when a field line forms a closed loop.
- B. The voltage of the neutral wire is always zero.
- C. AC power transmission losses are greater than DC losses.
- D. Fuse is an electrical safety device that operates to provide overcurrent protection of an electrical circuit.

So, statements A, B, and C are correct and statement D is incorrect.

22.

(c) No two magnetic lines of force can intersect each other.

Explanation: No two magnetic lines of force can intersect each other.

Section E

23.

(c) Silver

Explanation: Silver metal should be preferred for making the transmission line as its resistivity is lowest among the given metals. However, due to its higher cost, copper is preferred as its resistivity is also low.

Section F

24. (a) $12 \times 10^{-25} \text{ N}$

Explanation: $12 \times 10^{-25} \text{ N}$

Section G

25. When an electric current is passed through a metallic conductor, a magnetic field is set-up around it which exists so long as the current flows in the conductor. This is called magnetic effect of electric current.

26. i. Divergence or degree of closeness of magnetic field lines near the ends of a current-carrying straight solenoid indicates an increase in the strength of the magnetic field near the ends of the solenoid.
- ii. A current-carrying solenoid acts as a bar magnet. We know that a freely suspended bar magnet aligns itself in the North-South direction. So, a freely suspended current-carrying solenoid also aligns itself in the North-South direction.
- iii. Burnt out fuse cannot be re-used. Also, a fuse wire works because of its lower melting point. If the fuse with a larger rating is used with an appliance, the fuse wire shall not melt and hence would fail to serve the required purpose. So, a new fuse of the same rating should be used for electrical safety.

27. i. Yes, **alpha particles** are positively charged so they will induce current when flow and hence produce magnetic field around them.

ii. No, the neutrons being electrically neutral constitute no current.

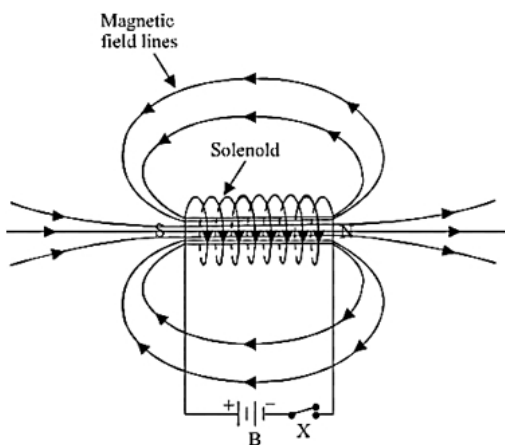
28. **Overloading:** When too many devices are connected to the same circuit and are operated at time, the wire draws a much large current (greater than its capacity) from mains is called overloading. This causes heating in the circuit that can lead to sparking. Causes of overloading:

- i. Live and neutral wires come in contact with each other overloading occurs.
- ii. Connecting too many appliances to a single socket.

Prevention of overloading

One should always install a fuse or Mini Circuit Breaker to every switchboard to prevent this situation. Proper earthing should be installed in the domestic electric circuit. If the earthing is appropriately done in a circuit, then the excess current will flow through the earthing to prevent the circuit's overload.

29. A solenoid is a long coil containing a large number of close turns of insulated copper wire. Magnetic field lines inside a current-carrying solenoid are uniform.



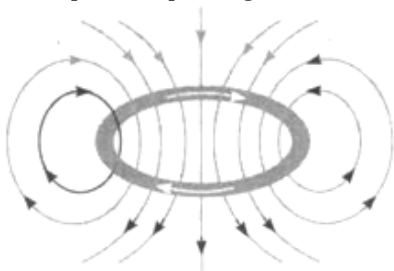
30. Fuse is used for protecting appliances due to short-circuiting or overloading. The fuse is rated for a certain maximum current and blows off when a current more than the rated value flows through it. If a fuse is replaced by one with larger ratings, the appliances may get damaged while the protecting fuse does not burn off. This practice of using fuse of improper rating should always be avoided.
31. At the axis of a current carrying circular loop, magnetic field lines are straight. Hence they appear closer to each other. When we move towards the periphery of current carrying circular loop, magnetic field lines turn so as to encircle the conductor. Due to this, they appear to be diverging.
32. The S.I unit of magnetic field is Tesla (T). The magnetic field strength is said to be one Tesla if one meter long conductor placed perpendicular to the direction of magnetic field ,carrying one ampere current experiences one newton of force.
33. The factors affecting the strength of a magnetic field at a point due to a straight current-carrying conductor are the magnitude of the electric current and the perpendicular distance between the point and the conductor. Right-hand thumb rule determines the direction of magnetic field produced in this case.
34. i. Total power of appliances used simultaneously
 $P = (3 \times 100) + (3 \times 40)$
 $= 420 \text{ W}$
 Voltage of mains, $V = 220\text{V}$
 Current drawn from the mains, $I = \frac{P}{V} = \frac{420}{220} = 1.91 \text{ A}$
- ii. Excess current available which can be safely used
 $= 5 \text{ A} - 1.91 \text{ A}$
 $= 3.09 \text{ A}$
 $\therefore$ Current drawn by each CFL of 40 W at 220 V,
 $I = \frac{P}{V} = \frac{40}{220} = 0.18 \text{ A}$
 $\therefore$ Number of additional CFLs of 40 W each which can be lighted,
 $n = \frac{\text{Excess current available}}{\text{Current drawn by each tube light}}$
 $= \frac{3.09}{0.18} = 17.2$
 $\therefore$ Seventeen additional CFLs can be lighted.
35. Total power of appliances used simultaneously $= (3 \times 100) + (8 \times 40)$
 $= 300 + 320$
 $= 620 \text{ W}$
 Voltage of mains, $V = 220 \text{ V}$
 Current drawn from mains, $I = \frac{P}{V} = \frac{620}{220} = 2.82 \text{ A}$
 Excess current available $= 5\text{A} - 2.82 \text{ A} = 2.18 \text{ A}$
 Current drawn by each of 60 W bulb at 220 V,
 $I = \frac{P}{V} = \frac{60}{220} = 0.27\text{A}$
 Number of additional bulbs of 60 W which can be lighted,
 $n = \frac{\text{Excess current available}}{\text{Current drawn by each bulb}}$
 $= \frac{2.18}{0.27} = 8.07$
 Therefore, eight more 60 W bulbs can be lighted.
36. (a) The maximum amount of current that can be passed through the fuse wire without melting it.
 (b) Copper or alloy of lead and tin.
 (c) Electric fuses and earth wire.

37. Here, $P = 3 \text{ kW} = 3000 \text{ W}$, $V = 220 \text{ V}$ Current drawn by motor,

$$I = \frac{P}{V}$$
$$= \frac{3000}{220} \text{ A}$$
$$= 13.6 \text{ A}$$

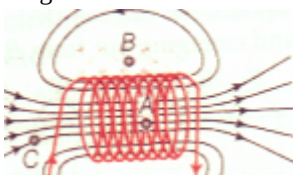
Hence, current rating of fuse to be used with this motor is 15 A.

38. i. 1. The force acting on a current-carrying conductor placed perpendicular to a magnetic field increases with the increase in the current flowing through a conductor.
2. When a horseshoe magnet is replaced by a stronger magnet, then the magnetic field increases. Thus, the force acting on the conductor increases.
3. If the length of the conductor increases then the force acting on the conductor also increases.
- ii. Fleming's left hand rule: Stretch the forefinger, middle finger and the thumb of your left hand mutually perpendicular to each other. If the forefinger indicates the direction of magnetic field and the middle finger indicates the direction of current, then the thumb will indicate the direction of motion of conductor.
39. Since the current passes through the loop in clockwise direction, therefore, the front face of the loop will be the south pole and the back face, i.e., the face touching the table will be north pole. According to right-hand rule, the direction of magnetic field inside the loop will be pointing downward. Outside the loop, the direction of the magnetic field will be upward.



40. i. When the key is closed, the galvanometer needle deflects momentarily in one direction.
- Reason:** When the key is closed, magnetic field lines around coil 2 increases momentarily. This causes an induced current to flow through it and hence deflection occurs in one direction.
- ii. When the key is opened, the galvanometer needle deflects again momentarily but in the opposite direction. Reason: When the key is open, magnetic field lines around coil 2 decreases momentarily. This causes an induced current to flow in the opposite direction. Hence, deflection occurs in the opposite direction.
41. The earth pin is made long so that the earth connect is made first and this ensures the safety of the user. If the appliance is defective, then as soon as the live pin gets connected, a strong current flow through the earth wire and the fuse blows off. The earth pin is thicker than the other two pins so that even by mistake it cannot be inserted in the hole for the live or neutral connection.
42. The direction in which a compass comes to rest in the magnetic field is called direction of line of force. No two magnetic lines of force cross each other because if they did so, it will mean that at the point of intersection, the compass needle would point towards two directions which is not possible. Hence two magnetic lines never intersect each other.
43. The lines drawn in a magnetic field along which north magnetic pole moves, are called magnetic field lines. The characteristic properties of magnetic field lines are:
- i. The magnetic lines originate from north pole and ends at south pole.
- ii. The magnetic lines do not intersect each other.
44. (a) A short circuit is simply a low resistance connection between the two conductors supplying electrical power to any circuit. Due to this huge amount of current flows through the circuit which in turn produces more heat which can cause fire.
- (b) Overloading means large amount of current flows in the circuit. It can happen when many electrical appliances of high power ratings are connected in a single socket.
- Overloading can be avoided by the following methods:**
- (i) Not supplying power to too many appliances through a single socket.
- (ii) To apply preventive methods of short circuiting.
45. (1) Magnetic field around a magnet.
- (2) Magnetic field around current carrying conductor.
- (3) Magnetic field around a current carrying solenoid.
- (4) Magnetic field around an electromagnet.

46. If sometimes a live wire touches neutral wire or earth wire, a large current flows through the circuit due to almost zero resistance of the circuit. This is called short-circuiting. To save the circuit from damage feared due to over-loading or short circuiting, a fuse of proper rating is put in each circuit.
47. a. The rods move if a current is sent through them from B to A. The force acting on the current carrying rod is what is responsible for this displacement. The rod is pulled to the left by the magnet, which causes a deflection of the rod in that direction.
b. No change is observed when the axis of the rod "AB" is moved and aligned parallel to the magnetic field and current is passed in the rod in the same direction. Since there is no force, so there will be no motion.
48. i. As the distance from conductor increases, the strength of the magnetic field decreases i.e. $B \propto \frac{1}{r}$.
ii. As the magnitude of the current flowing through the conductor increases, the strength of magnetic field increases, i.e. $B \propto I$.
49. The magnetic field lines of force are the lines drawn in a magnetic field along which a hypothetical north magnetic pole would move if it is free to do so.
- The direction of a magnetic field at a point is the direction of the resultant force acting on a hypothetical north pole placed at that point. The tangent at any point on the magnetic field line gives the direction of magnetic field at that point. The direction of the magnetic field at a point can be found by placing a small magnetic compass at that point. The north end of the needle indicates the direction of the field.
- Two important properties of the magnetic lines of force are:
- (i) The magnetic lines of force start from north pole and terminate at south pole. Inside the magnet they travel from south pole to north pole. Thus, they are closed curves.
(ii) They do not intersect each other because at the point of intersection there will be two directions of same magnetic field which is impossible.
50. Force on electron is maximum in fig. (a) because here direction of motion of electron is at right angles to that of magnetic field 'B'. The force is minimum (or zero) in fig. (c) because here electron is moving along the direction of magnetic field B.
The direction of maximum force acting on electron is perpendicular to the plane of paper and directed into it.
51. In case of movement of a charged particle, the magnetic field is created around the path on which the charged particle moves
i. Yes, because α -particles are charged particles so when they move they create a magnetic field around it.
ii. No, as neutrons carry zero charges, so no magnetic field would be produced around its path.
52. Fleming's left hand rule states that stretch the forefinger, the central finger and the thumb of your left hand mutually perpendicular to each other. If the forefinger shows the direction of the magnetic field and central finger that of the current, then the thumb will point towards the direction of motion of the conductor.
53. The displacement of the conductor
(i) will increase on increasing the current
(ii) Will decrease on using a weak horse shoe magnet.
54. Magnetic field lines due to a solenoid



In case of an ideal solenoid, magnetic field strength is maximum at point A and is minimum or zero at point B. This is because the magnetic field is strong, where magnetic field lines are far apart. At the point, C, the density of the field lines is less than that of point A, but greater than that of point B. So, the order of the magnetic field at points A, B and C is $B_B < B_C < B_A$.

55. **Hazards of electricity** : Electricity is the most important source of energy. Whereas a proper use of this is a boon, improper use is very dangerous and may prove to be fatal. There are number of hazards.
(1) Loose connections, defective switches and sockets can cause sparking which may lead to fire.
(2) If a person touches a live wire, he gets a very severe shock. This may prove fatal.
(3) Short-circuiting due to damaged wires or overloading of circuits can cause fires.
56. i. The magnitude of force acting on a current carrying conductor placed in a magnetic field will be zero, when the current carrying conductor is in the direction of magnetic field.
ii. The magnitude of force acting on a current carrying conductor placed in a magnetic field will be maximum, when the current carrying conductor is normal (perpendicular) to the magnetic field.
57. i. Long transmission distances of alternating current can be achieved with little loss of electrical energy. As a result, it is preferable to direct current for long-distance electric energy transmission.

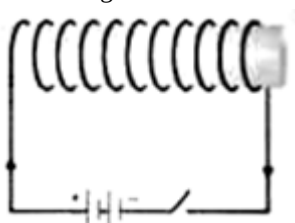
- ii. Direct Current, or DC, is supplied by a dry cell (or battery of any kind). Alternating Current, or AC, is supplied by household outlets.
- iii. The fuse used in electrical circuits is a typical example of Joule's heating in action. It safeguards appliances from excessively high electric current. It is composed of a wire made of a metal or alloy such as lead, copper, iron, or aluminium. The temperature of the fuse wire rises if a current greater than the allowed amount flows through the circuit. The fuse wire melts as a result, breaking the circuit. Typically, the fuse wire is enclosed in a porcelain or similar material cartridge with metal ends.
58. A uniform magnetic field in a region is represented by drawing parallel and equidistant straight lines, all pointing in the same direction as shown in figure:



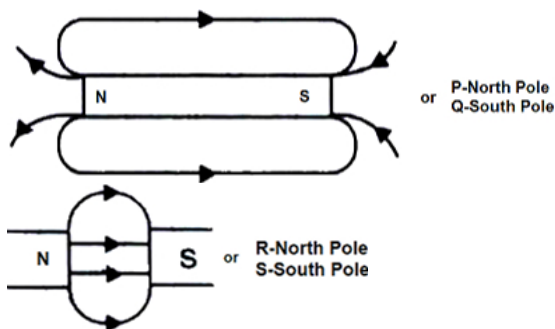
59. A current carrying coil contains charged particles which experiences a force ($F=Bqv$). The total force experienced by the charged particle is equal to the force experienced by the conductor which is perpendicular to both the magnetic field and the direction of current in the conductor.
60. a. **Fleming's left-hand rule:** Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.
b. South
61. **Short circuiting :** If sometimes a live wire touches neutral wire or earth wire, a large current flows through the circuit due to almost zero resistance of the circuit. This is called short-circuiting. To save the circuit from damage feared due to over-loading a short-circuiting, a fuse of proper rating is put in each circuit.
- Over loading :** The supply wires as well the wires used in household wiring has a specific rating. The rating of 15 A means that if a current upto 15 A is passed through circuit, there is no likely damaged feared to the circuit. But if current more than maximum allowed limit is passed, there may be excessive heating of the wires and it may damage the wiring due to excessive heating.
62. Here, Power, $P = 3 \text{ kW}$
 $= 3 \times 1000 \text{ W} = 3000 \text{ W}$
 Supply voltage, $V = 220 \text{ V}$
 $\therefore$ Current drawn, $I = \frac{P}{V} = \frac{3000}{220}$
 $= 13.6 \text{ A}$
 Since, current rating of fuse is 15 A, hence the fuse can be used with the electric heater.

63. Permanent Magnet	Electromagnet
(i) Very strong electromagnets cannot be produced.	(i) Very strong electromagnets can be produced.
(ii) Its strength is fixed.	(ii) Strength can be changed by changing the current through coil.
(iii) Polarities are fixed.	(iii) Polarities can be reversed by changing the direction of the current.
(iv) Cannot be immediately demagnetized.	(iv) Can be demagnetized immediately by stopping the current in the coil.
(v) Example of permanent magnet is a Bar Magnet	(v) Example of a temporary magnet is solenoid wound across a nail and connected to a battery.

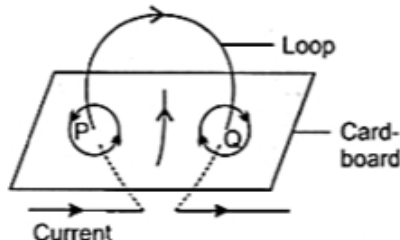
64. The magnetic field produced due to current flowing in a coil or a solenoid can be used to magnetise a material like soft iron temporarily. The insulated copper wire is wrapped on a soft iron piece. When current is passed through the coil using a battery and closing a key the iron piece behaves like a bar magnet as long as current is being passed. Such a magnet is called an electromagnet.



65. A compass needle is itself a magnet and when placed near a magnet gets deflected due to the repulsive magnetic force between like poles and attractive magnetic force between unlike force of the magnet.
66. A current carrying solenoid behaves like a bar magnet. It rest in geographic north-south direction.
67. The magnetic field lines emerge from the North-pole and merge to South-pole outside the magnet. (Inside the magnet the direction is from South pole to North pole.)



68. Figure shows the magnetic lines of force due to current Carrying Coil.



The magnitude of magnetic field at the centre of coil depend on:

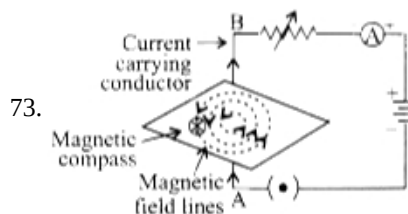
- the strength of current in the coil, and
- the number of turns in the coil.

69. Earth will acts as a safety device. When live wire touches the metallic appliance then electric current flows through casing to the earth instead of human body and thus we prevent ourselves from getting shock. It is necessary to earth metallic casing of the appliance because it saves electrical appliance from burning and prevent us from the electric shock.

70. Strength of the magnetic field falls as distance increases. This is indicated by the decrease in degree of closeness of the lines of field.

71. The magnetic field generated in the centre, or core, of a current carrying solenoid is essentially uniform, and is directed along the axis of the solenoid. Outside the solenoid, the magnetic field is far weaker.

72. The thumb in Right-Hand Thumb Rule indicates the direction of current flowing in a straight conductor. Comparing with Fleming's Left-Hand Rule, in which the thumb indicates the direction of force exerted by the magnetic field developed around the current carrying conductor.



- The strength of a magnetic field is inversely proportional to the square of the distance from the conductor i.e., strength of an electric field decreases with increase in distance.
- The strength of the magnetic field is directly proportional to the current passing in the wire i.e., strength of the magnetic field increases with the increase in current.

74. a. A coil of insulated copper wire wound in the form of a cylinder is a solenoid. When current is passed through a solenoid, it produces magnetic field lines like a bar magnet.

- The pattern of magnetic field lines of a solenoid is shown in the figure. Inside the solenoid, field lines are parallel to each other This indicates that the magnetic field is uniform .

